the nutrients, water and electrolytes from the external environment into every cell in the body through the circulatory system.

5.1 Digestive system

The process of digestion involves intake of the food (Ingestion), breakdown of the food into micromolecules (Digestion), absorption of these molecules into the blood stream (Absorption), the absorbed substances becoming components of cells (Assimilation) and elimination of the undigested substances (Egestion). Digestive system includes the alimentary canal and associated digestive glands.

5.1.1 Structure of the alimentary canal

The alimentary canal is a continuous, muscular digestive tract that begins with an anterior opening, the mouth and opens out posteriorly through the anus. The alimentary canal consists of mouth, buccal cavity, pharynx, oesophagus, stomach, intestine, rectum and anus (Figure. 5.1). The mouth is concerned with the reception of food and leads to the buccal cavity or oral cavity (Figure. 5.2). Mechanical digestion is initiated in the buccal cavity by chewing with the help of teeth and tongue. Chemical digestion is through salivary enzymes secreted by the salivary glands.

Each tooth is embedded in a socket in the jaw bone this type of attachment is called **thecodont**. Human beings and many mammals form two sets of teeth during their life time, a set of 20 temporary milk teeth (deciduous teeth) which gets replaced by a set of 32 permanent teeth (adult teeth). This type of dentition is called **diphyodont**. The permanent teeth are of four different types (**heterodont**), namely, Incisors (I) chisel like cutting teeth, Canines (C) dagger shaped tearing teeth, Pre molars (PM) for grinding, and Molars (M) for grinding and crushing. Arrangement of teeth in each half of the upper and lower jaw, in the order of I, C, PM and M can be represented by a dental formula, in human the dental formula is

$$\frac{2123}{2123} \times 2$$

Mineral salts like calcium and magnesium are deposited on the teeth and form a hard layer of '**tartar' or calculus** called plaque. If the plaque formed on teeth is not removed regularly, it would spread down the tooth into the narrow gap between the gums and enamel and causes inflammation called **gingivitis**, which leads to redness and bleeding of the gums and to bad smell. The hard chewing surface of the teeth is made of enamel and helps in mastication of food.

Tongue is a freely movable muscular organ attached at the posterior end by the frenulum to the floor of the buccal cavity and is free in the front. It acts as a universal tooth brush and helps in intake food, chew and mix food with saliva, to swallow food and also to speak. The upper surface of the tongue has small projections called papillae with taste buds.

The oral cavity leads into a short common passage for food and air called pharynx. The oesophagus and the trachea (wind pipe) open into the pharynx. Food passes into the oesophagus through a wide opening called gullet at the back of the pharynx. A cartilaginous flap called epiglottis prevents the entry of food into the glottis (opening of trachea) during

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